

As a reminder the matrix are :

$${}^0T_1 = \begin{pmatrix} c_{\alpha} & c_{\beta} & c_{\gamma} & p(\alpha) \\ s_{\alpha} & s_{\beta} & s_{\gamma} & p(\beta) \\ c_{\alpha} & c_{\beta} & c_{\gamma} & p(\gamma) \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{aligned} {}^1A_2 &= c_{\alpha}c_{\beta}c_{\gamma}c_{\delta} - s_{\alpha}s_{\beta}c_{\delta} - s_{\alpha}c_{\beta}c_{\delta} - c_{\alpha}s_{\beta}c_{\delta} + s_{\alpha}s_{\beta}c_{\delta} \\ {}^1A_3 &= c_{\alpha}(-c_{\beta}c_{\gamma}c_{\delta} - s_{\alpha}s_{\beta}c_{\delta}) - s_{\alpha}(s_{\beta}c_{\delta}) = -c_{\alpha}c_{\beta}c_{\gamma}c_{\delta} - c_{\alpha}s_{\beta}c_{\gamma}c_{\delta} - s_{\alpha}s_{\beta}c_{\delta} \\ {}^1A_4 &= c_{\alpha}c_{\beta}c_{\gamma}c_{\delta} + s_{\alpha}s_{\beta}c_{\delta} + s_{\alpha}c_{\beta}c_{\delta} + c_{\alpha}s_{\beta}c_{\delta} \\ {}^1A_5 &= c_{\alpha}(c_{\beta}c_{\gamma}c_{\delta} + s_{\alpha}s_{\beta}c_{\delta} - d_{\alpha}s_{\beta}c_{\gamma}c_{\delta} - d_{\alpha}s_{\beta}c_{\gamma}c_{\delta}) + s_{\alpha}d_{\alpha} + c_{\alpha}d_{\alpha} \\ {}^1A_6 &= s_{\alpha}c_{\beta}c_{\gamma}c_{\delta} - s_{\alpha}s_{\beta}c_{\delta} - c_{\alpha}s_{\beta}c_{\delta} \\ {}^1A_7 &= -s_{\alpha}c_{\beta}c_{\gamma}c_{\delta} + c_{\alpha}s_{\beta}c_{\delta} \\ {}^1A_8 &= -s_{\alpha}s_{\beta}c_{\gamma}c_{\delta} - c_{\alpha}c_{\beta}c_{\delta} \\ {}^1A_9 &= s_{\alpha}(c_{\beta}c_{\gamma}c_{\delta} + s_{\alpha}s_{\beta}c_{\delta} - d_{\alpha}s_{\beta}c_{\gamma}c_{\delta} - d_{\alpha}s_{\beta}c_{\gamma}c_{\delta}) - c_{\alpha}d_{\alpha} - c_{\alpha}d_{\alpha} \\ {}^1A_{10} &= s_{\alpha}c_{\beta}c_{\gamma}c_{\delta} + c_{\alpha}s_{\beta}c_{\delta} + c_{\alpha}s_{\beta}c_{\delta} \\ {}^1A_{11} &= -s_{\alpha}s_{\beta}c_{\gamma}c_{\delta} - c_{\alpha}c_{\beta}c_{\delta} \\ {}^1A_{12} &= -s_{\alpha}s_{\beta}c_{\gamma}c_{\delta} \\ {}^1A_{13} &= c_{\alpha}s_{\beta}c_{\gamma}c_{\delta} + c_{\alpha}s_{\beta}c_{\delta} - d_{\alpha}s_{\beta}c_{\gamma}c_{\delta} - d_{\alpha}s_{\beta}c_{\gamma}c_{\delta} \end{aligned}$$

So following Z-X-X Euler angles we have

$$\beta = \arctan2(\sqrt{c_{\alpha}^2 + s_{\alpha}^2}, c_{\alpha})$$

$$\alpha = \arctan2(c_{\beta}, s_{\beta})$$

$$\gamma = \arctan2(c_{\gamma}, s_{\gamma})$$

These is a singularity if $|c_{\alpha}| = 1$

In this case we take $\gamma = 0$ and $\alpha = \arctan2(c_{\beta}, s_{\beta})$

And the inverse kinematic gave us:

$$\begin{cases} \Delta < 0 & \text{no solution } (c_i \in [-1, 1]) \\ \Delta = 0 & \theta_i = \pm \arccos\left(\frac{-B}{2A}\right) \\ \Delta > 0 & \theta_i = \pm \arccos\left(\frac{-B \pm \sqrt{\Delta}}{2A}\right) \end{cases}$$

$$\theta_1 = \arctan2\left(\pm\sqrt{1 - \left(\frac{-d_1 - c_1 d_2}{r}\right)^2}, \frac{-d_1 - c_1 d_2}{r}\right) + \arctan2(-p_x, p_y)$$

$$\theta_2 = \arctan2\left(\frac{1}{s_1}(c_1 r_{1,1} - s_1 r_{1,2}), \frac{1}{s_1}(s_1 r_{1,1} - c_1 r_{1,2})\right)$$

After obtaining θ_1 we have:

$$\theta_2 = \arctan2(k_1 k_3 - k_1 k_4, k_1 k_3 + k_2 k_4)$$

$$\theta_3 = \arctan2\left(\pm\sqrt{1 - \left(\frac{1}{2a_{1,3}}(k_1^2 + k_2^2 - k_3^2 - a_1^2)\right)^2}, \frac{1}{2a_{1,3}}(k_1^2 + k_2^2 - k_3^2 - a_1^2)\right) + \arctan2(-k_6, k_5)$$

For this motion planning problem multiple choices are made:

-The first and most important one is the usage of joint space.

This space prevents the appearance of singularities and so highly simplify the problem.

It also drastically reduce the calculation time as inverse kinematic in our case is time consuming.

While reducing the computation of inverse kinematics it also imply reducing the amount of choices needed in case of multiple solutions.

-The second choice is the method of selection in case of multiple solutions of the inverse-kinematic.

To make this choice let's define : $C(\theta, \theta') = \sum_{i=1}^n (x_i - x'_i)^2$.

Here the idea is to use C as a cost function to estimate the cost of a movement from θ to θ' .

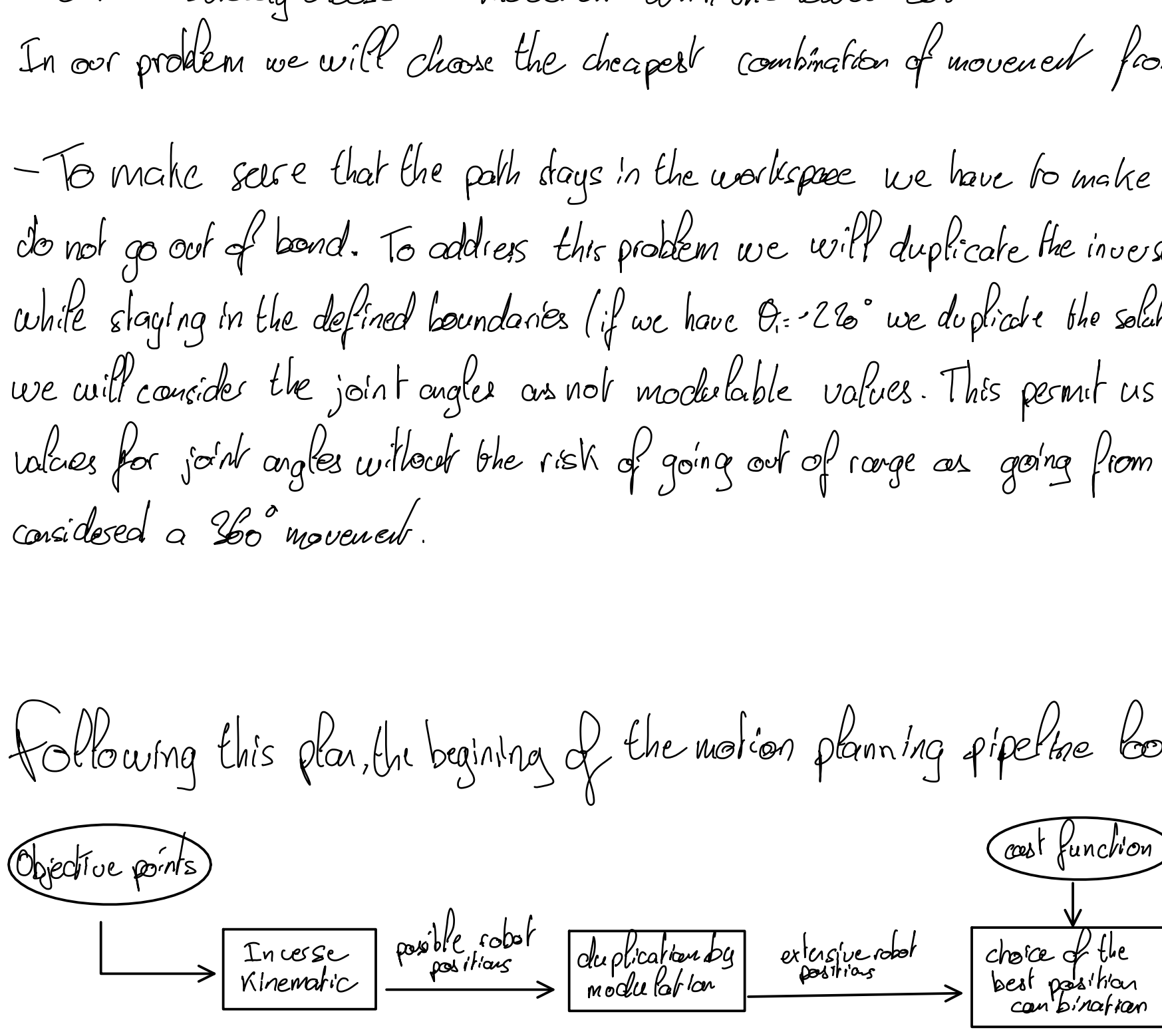
θ and θ' being vectors of joint angles. The cost here is a sort of weighted distance making a movement of the base more important than one of the wrist.

We then obviously choose the movement with the lowest cost.

In our problem we will choose the cheapest combination of movement from A to B and then C.

-To make sure that the path stays in the workspace we have to make sure that the joint angles do not go out of bound. To address this problem we will duplicate the inverse kinematic solutions using modulus while staying in the defined boundaries (if we have $\theta_i = 230^\circ$ we duplicate the solution to $\theta_i = 110^\circ$). After this step we will consider the joint angles as not modifiable values. This permit us to have all possible modulus values for joint angles without the risk of going out of range as going from $\theta_i = 230$ to $\theta_i = 110$ will be considered a 360° movement.

Following this plan, the beginning of the motion planning pipeline looks something like this:



Next is the actual motion planning.

We know $t_{acc} = 0.5 \text{ sec}$: half transition time

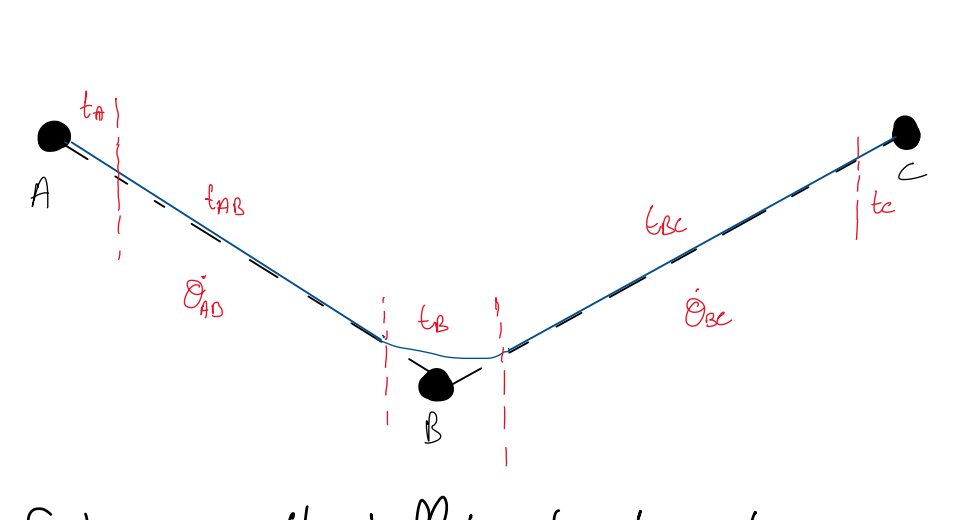
$t_{AB} = 5.0 \text{ sec}$: overall duration from A to B

$t_{BC} = 4.0 \text{ sec}$: overall duration from B to C

for the path computation we will use a sampling time of 0.01 sec .

Let's call $(\theta_A, \theta_B, \theta_C) \in \mathbb{R}^6$ the chosen positions for each point.

The problem is now to find the red unknown using the values defined just before.



First we know the half transition time t_{acc} , so we know $t_A = t_C = t_{acc}$ and $t_B = 2t_{acc}$

Furthermore $t_{AB} = t_{AB} - t_A - \frac{1}{2}t_B = t_{AB} - t_{acc}$

$t_{BC} = t_{BC} - \frac{1}{2}t_B - t_C = t_{BC} - 2t_{acc}$

$$\begin{aligned} \theta_B &= \frac{\theta_C - \theta_A}{t_{BC} - \frac{1}{2}t_B} = \frac{\theta_C - \theta_A}{t_{BC} - t_{acc}} \\ \theta_{AB} &= \frac{\theta_B - \theta_A}{t_{AB} - t_{acc}} \end{aligned}$$

Now we need the acceleration during the different transition.

We will consider the accelerations constant.

$\ddot{\theta}_A = 0$ and $\ddot{\theta}_B$ is continuous.

So $\ddot{\theta}_A = \ddot{\theta}_B = 0$. $\ddot{\theta}_{AB} = \ddot{\theta}(t_A)$

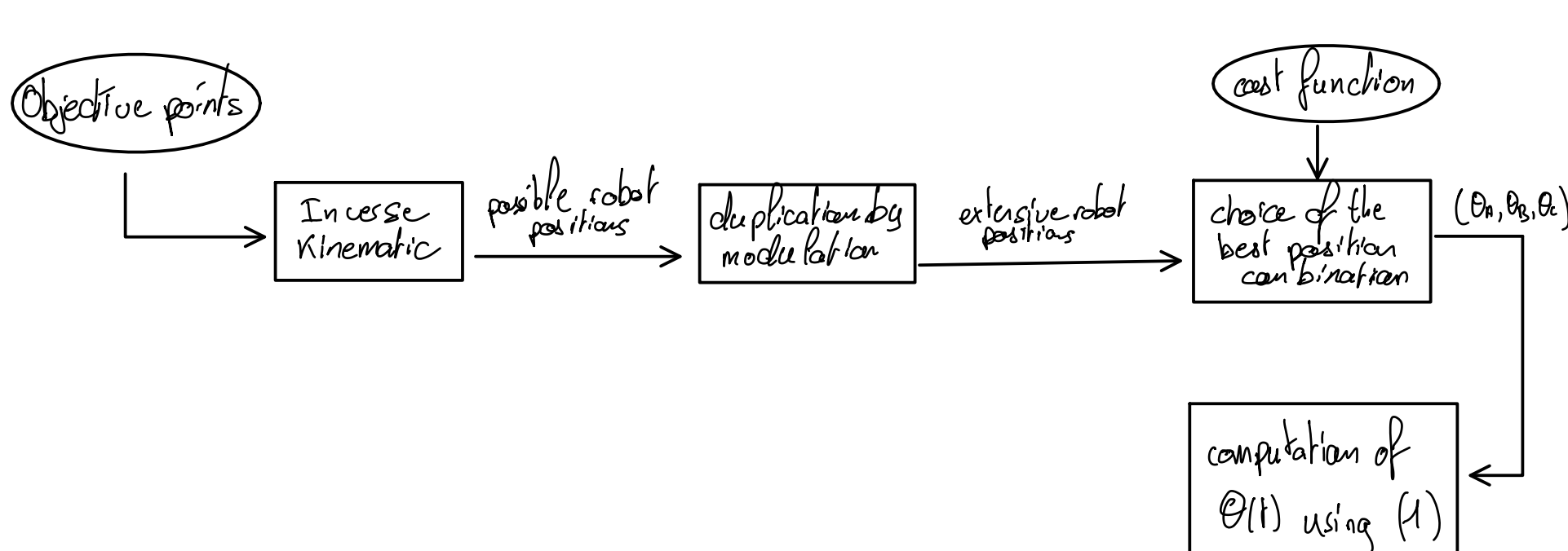
$$\text{And } \ddot{\theta}_A = \frac{\ddot{\theta}_{AB}}{t_{AB}}$$

Using the same logic $\ddot{\theta}_C = \frac{\ddot{\theta}_{BC}}{t_{BC}}$

$$\text{And } \ddot{\theta}_B = \frac{\ddot{\theta}_{BC} - \ddot{\theta}_{AB}}{2t_{acc}}$$

$$\theta(t) = \begin{cases} t \leq t_A, t_A \leq t \leq t_B \\ t_B \leq t \leq t_{AB} + t_B, \theta(t_A) + (t - t_A)\ddot{\theta}_{AB} \\ t_{AB} + t_B \leq t \leq t_{AB} + t_{BC} + t_B, \theta(t_{AB} + t_B) + (t - t_{AB} - t_B)\ddot{\theta}_{AB} \\ t_{AB} + t_B \leq t \leq t_{AB} + t_{BC} + t_B, \theta(t_{AB} + t_B) + (t - t_{AB} - t_B)\ddot{\theta}_{BC} \\ t_{AB} + t_B + t_{BC} + t_B \leq t, \theta(t_{AB} + t_B + t_{BC} + t_B) + (t - t_{AB} - t_B - t_{BC} - t_B)\ddot{\theta}_{BC} \end{cases} \quad (1)$$

Finally the complete pipeline is:



Results:

